

Advanced RAG Series

Powerful RAG with Hybrid Search

Keyword + Vector Search + Ensemble Retrieval

by Mayank Chugh



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BM25 / TF-IDF

Dense Embeddings

Ensemble Retriever

What is Hybrid Search?

Hybrid Search = Keyword Search (Sparse Vectors) + Vector Search (Dense Vectors)
combined via an Ensemble Retriever to achieve better retrieval quality.

Keyword Search

- Based on TF-IDF or BM25 algorithm
- Produces SPARSE vectors (mostly zeros)
- Excellent for exact-match terms
- Old but proven — used by Google for decades

Vector Search

- Uses Transformer-based embedding models
- Produces DENSE vectors (all values populated)
- Captures semantic / conceptual similarity
- Handles paraphrasing and synonyms well

The Problem with Naive Retrieval

01

Naive Retrieval

Standard similarity search — what most tutorials show

02

Hybrid Search

Keyword + Vector — THIS VIDEO



03

Sentence Window Retrieval

Fetches surrounding context window

04

Self-Query Retrieval

Model generates its own sub-queries

05

Parent Document Retrieval

Small-to-big chunking strategy

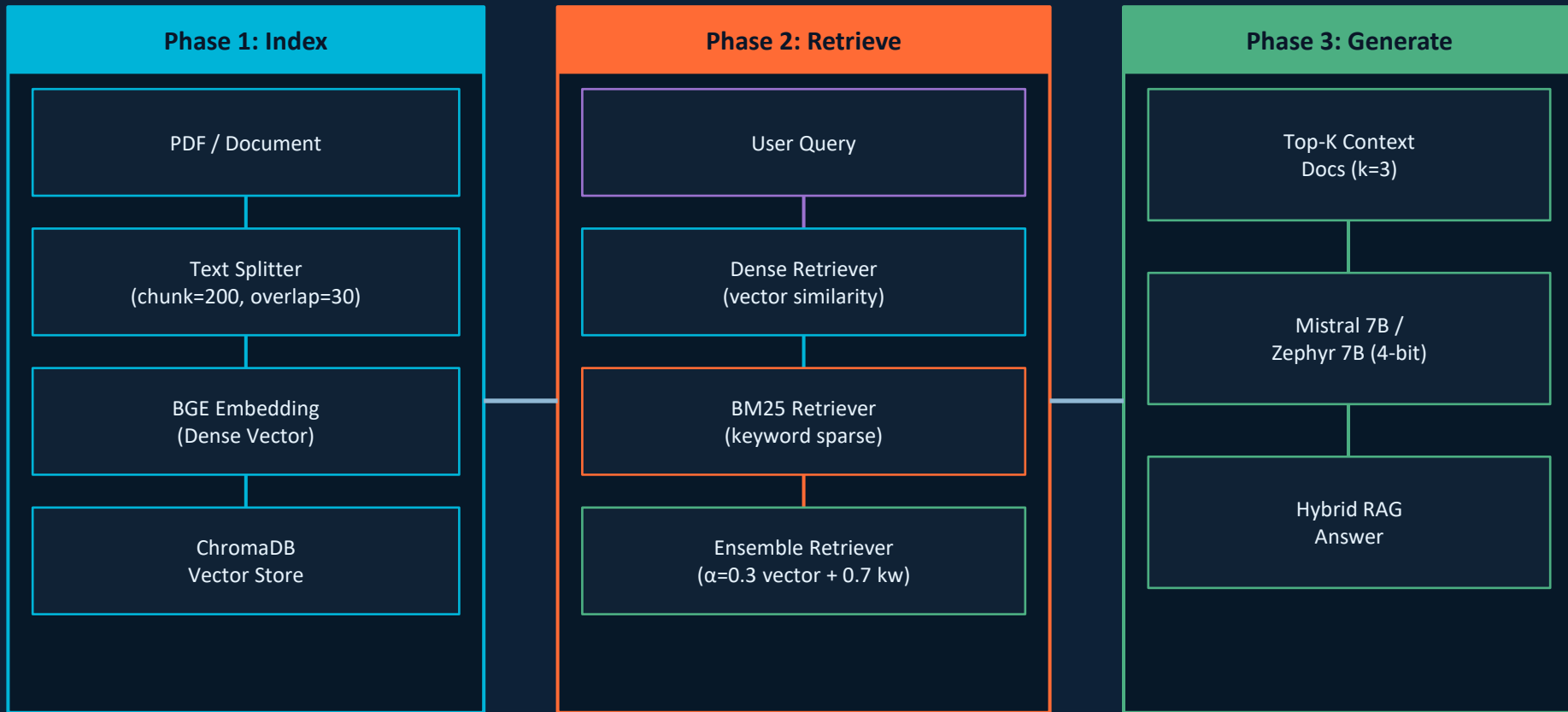
06

HyDE

Hypothetical Document Embeddings

Teaching Note: Naive retrieval is the baseline — it works but misses cases where exact words and semantic meaning both matter.

Hybrid RAG Architecture



Teaching Note: The key difference from naive RAG is Phase 2 — two parallel retrievers whose scores are blended by alpha weighting.

Keyword Search: TF-IDF & BM25

TF-IDF (Classic Keyword)

- **TF = Term Frequency**
- How often a word appears in THIS document
- **IDF = Inverse Document Frequency**
- How rare the word is across ALL documents
- **Score = $TF \times IDF$ per word in vocabulary**
- Result: SPARSE vector (mostly zeros)
- Pre-processing: lowercase + remove punctuation

BM25 (Updated Standard)

- Evolved version of TF-IDF
- Adds document length normalization
- Saturates term frequency (prevents spamming)
- Industry standard — Elasticsearch uses it
- Available via LangChain: `BM25Retriever`
- Also on HuggingFace as a sparse model
- Used in this demo for keyword retrieval

Analogy: The Library Search Desk

*Query: "My name is Sunny" — search for "Sunny" → found. Search for "Bunny" → not found.
But as a human you can see Bunny is phonetically close to Sunny. Math needs proof.*

Librarian A — Keyword Expert

- Has a card index of every exact word
- Looks up "Bunny" — not in index — returns nothing
- Brilliant at finding exact term matches
- Fails completely on paraphrases or misspellings

Librarian B — Semantics Expert

- Has read every book and understands concepts
- "Bunny" → thinks of small animals → finds rabbit books
- Brilliant at synonyms and related concepts
- May miss very specific technical search terms

Hybrid: Both Librarians Consult

- Both work in parallel on the same query
- A scores each book, B scores each book
- Scores are blended using alpha weighting
- Final ranked list is far more reliable



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Ensemble Score & Alpha Weighting

$$\text{Hybrid Score} = \alpha \times \text{Dense Score} + (1 - \alpha) \times \text{Sparse Score}$$

$\alpha = 1.0$

Pure Vector Search only

Dense: 100% | Sparse: 0%

Best for conceptual / semantic queries

$\alpha = 0.5$

Equal blend of both

Dense: 50% | Sparse: 50%

Balanced queries — good default

$\alpha = 0.3$

Keyword-dominant

Dense: 30% | Sparse: 70%

Technical docs with precise terminology

$\alpha = 0.0$

Pure Keyword Search only

Dense: 0% | Sparse: 100%

Exact-match search requirement

`weights=[0.3, 0.7] → 30% vector + 70% keyword → demo config in this video`

Teaching Note: Alpha is your tuning knob. For keyword-heavy domains like legal or medical, bias toward lower alpha. For conversational Q&A, higher alpha works better.

Hybrid vs Naive: Real Results

Query: "What is the RAG token model?" — Research Paper: Retrieval-Augmented Generation for NLP

Naive RAG (Dense Only)

Docs Retrieved:	3 retrieved documents
Context Quality:	Loosely related context
Answer:	Vague, mixed answer about general RAG
Verdict:	 Imprecise

Hybrid RAG (Dense + BM25)

Docs Retrieved:	6 retrieved documents (3+3)
Context Quality:	Precise + conceptually rich context
Answer:	RAG token model predicts each output token from a separate document from the retrieval corpus — accurate and concise
Verdict:	 Accurate & Concise



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Key Takeaways & What's Next

01 Naive retrieval uses only dense (semantic) vectors — it misses exact keyword matches.

02 TF-IDF and BM25 create sparse keyword vectors — powerful for precise technical terms.

03 Ensemble Retriever blends both scores via alpha weighting — fully tunable per domain.

04 Hybrid RAG delivers richer context to your LLM — resulting in more accurate answers.

05 Implementation in LangChain: BM25Retriever + ChromaDB retriever → EnsembleRetriever.



SOON

Next Episode: Hybrid Search with Weaviate + Cross-Encoder Reranking



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